

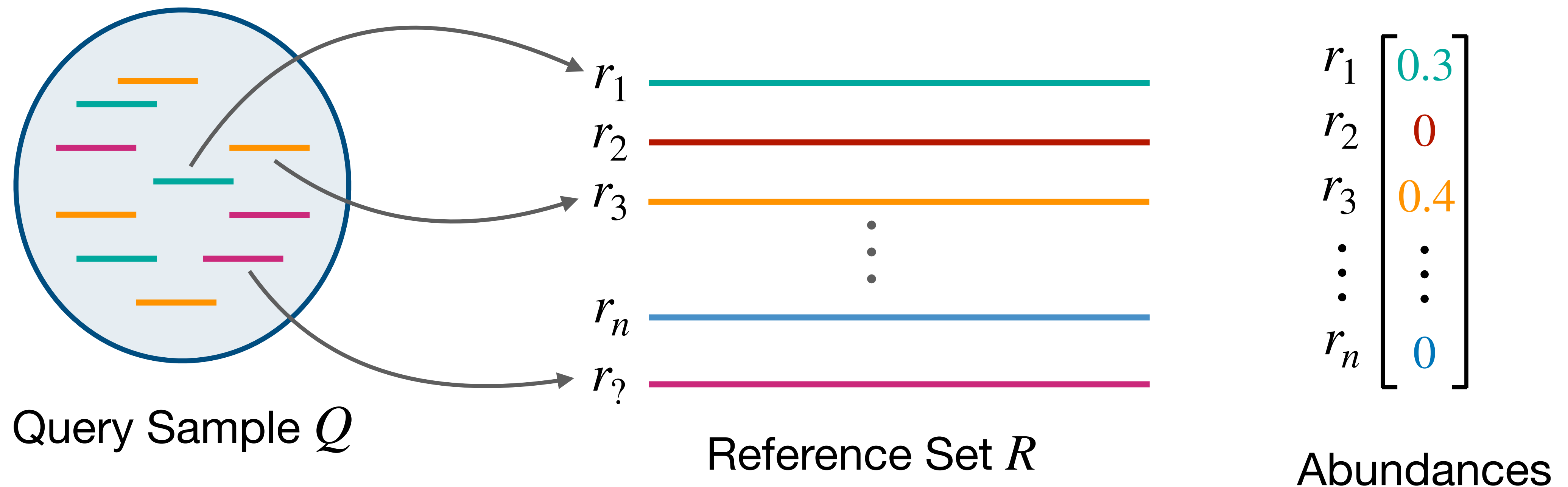
Tutorial on DecoDiPhy: Deconvolving Phylogenetic Distance Mixtures

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Metagenomic Profiling

- Metagenomic Profiling is the task of identifying the taxa in a query sample Q .

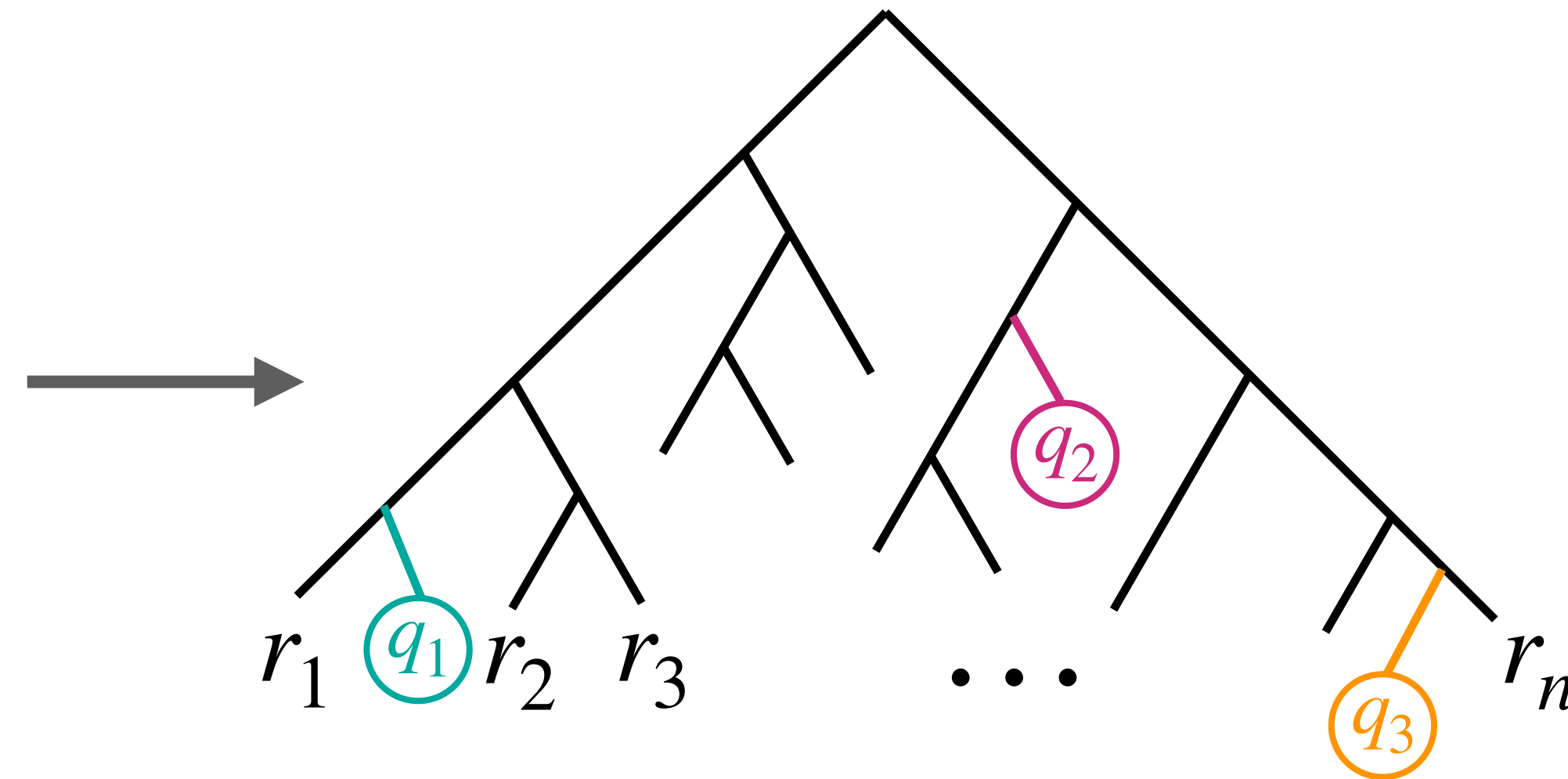


Metagenomic Profiling

- Phylogenetic placement allows for placement of all query taxa on the tree.



Query Sample Q



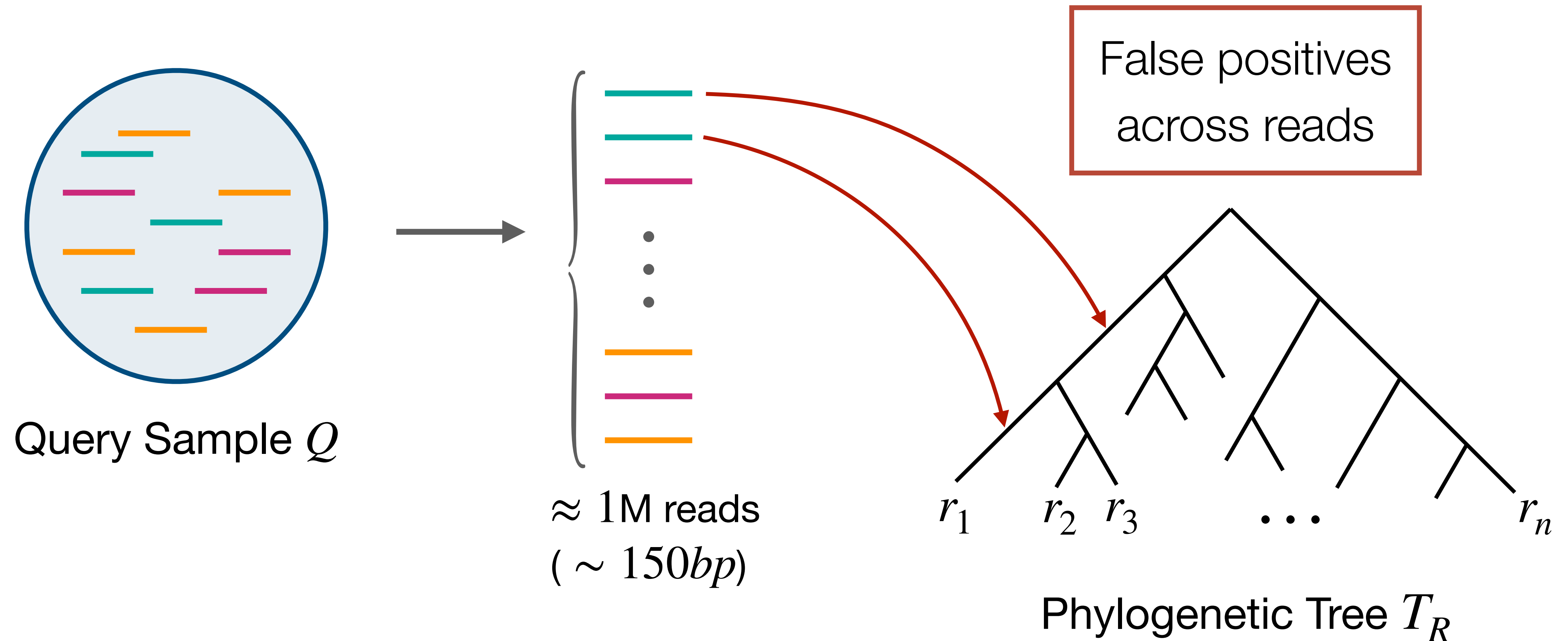
Phylogenetic Tree T_R

$$\begin{matrix} q_1 \\ q_2 \\ q_3 \end{matrix} \begin{bmatrix} 0.3 \\ 0.4 \\ 0.3 \end{bmatrix}$$

Abundances

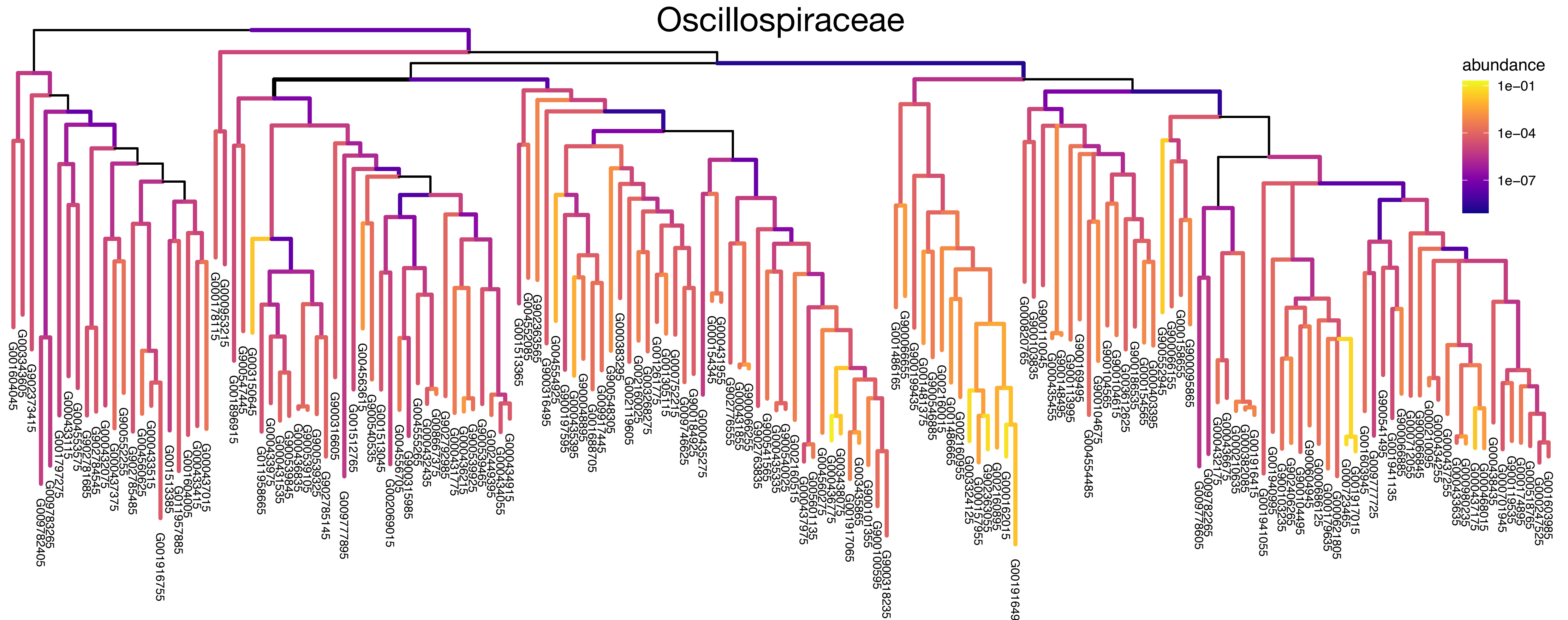
1. Read-level Classification

- Woltka (Zhu et al. 2022), krepp (Osman Berk Şapcı and Mirarab 2025)



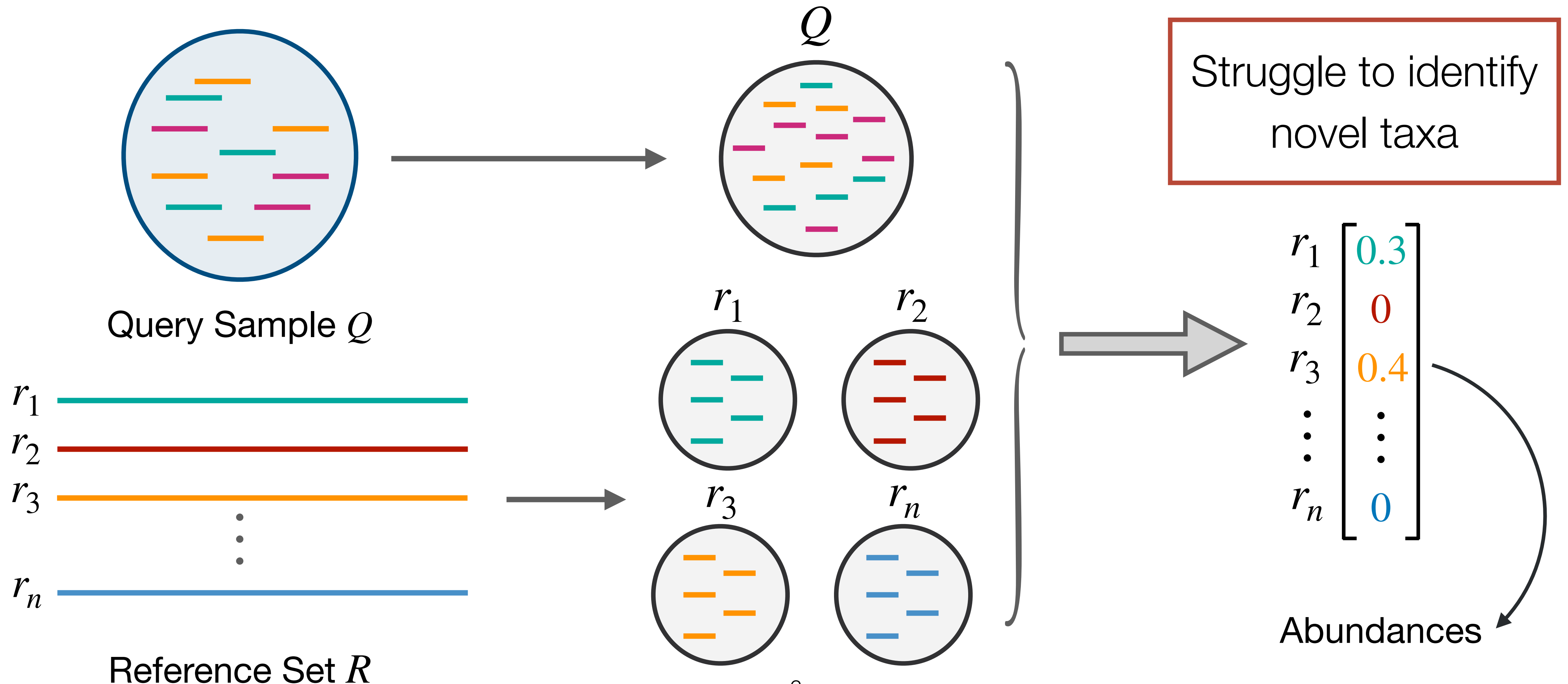
1. Read-level Classification (krepp)

- Inflammatory Bowel Disease (IBD) sample from Franzosa et al. 2019

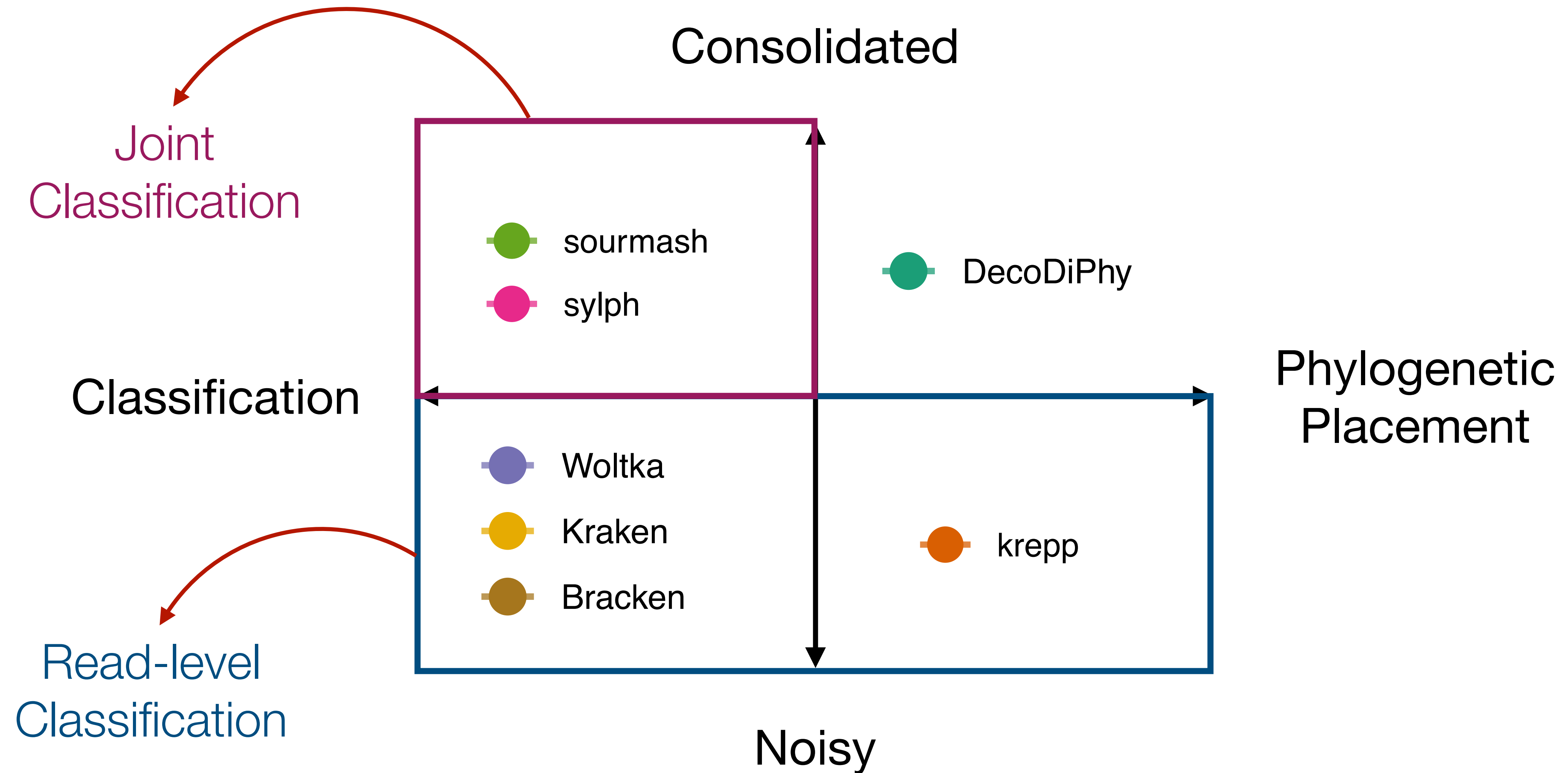


2. Joint Abundance Estimation

- Sourmash (Irber et al. 2022) and sylph (Shaw and Yu 2024)



Spectrum of Metagenomic Classification Tools



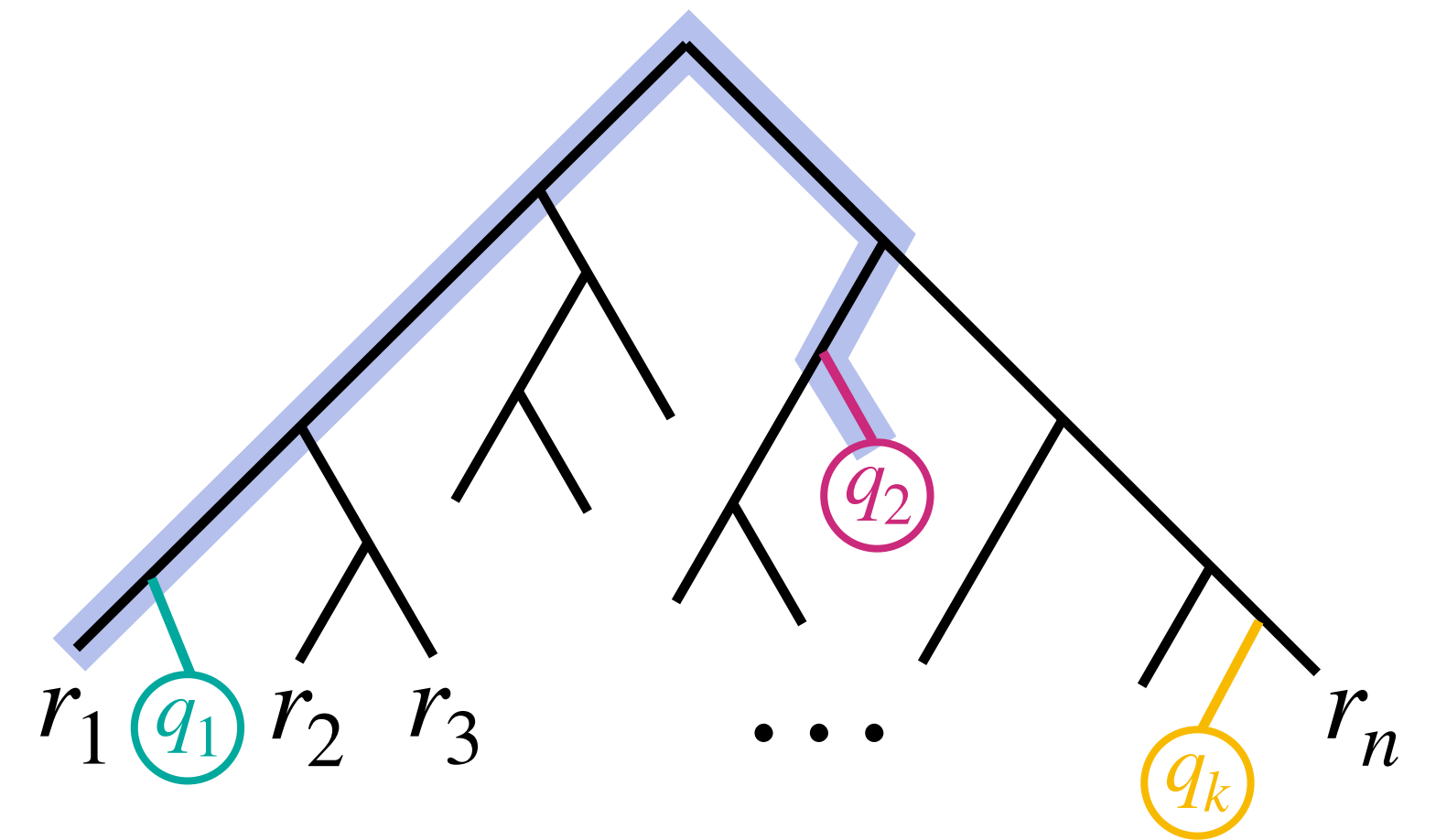
Mixture Distances (Definition)

Average
Distance Vector

Abundances

$$d = \begin{bmatrix} d(Q, r_1) \\ d(Q, r_2) \\ \vdots \\ d(Q, r_n) \end{bmatrix} = \begin{bmatrix} d(q_1, r_1) & d(q_2, r_1) & \dots & d(q_k, r_1) \\ d(q_1, r_2) & & & \\ \vdots & \ddots & & \vdots \\ d(q_1, r_n) & d(q_2, r_n) & \dots & d(q_k, r_n) \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_k \end{bmatrix}$$

$d \in \mathbb{R}^n$ $D_{T^*} \in \mathbb{R}^{n \times k}$ $p \in \Delta^{k-1}$



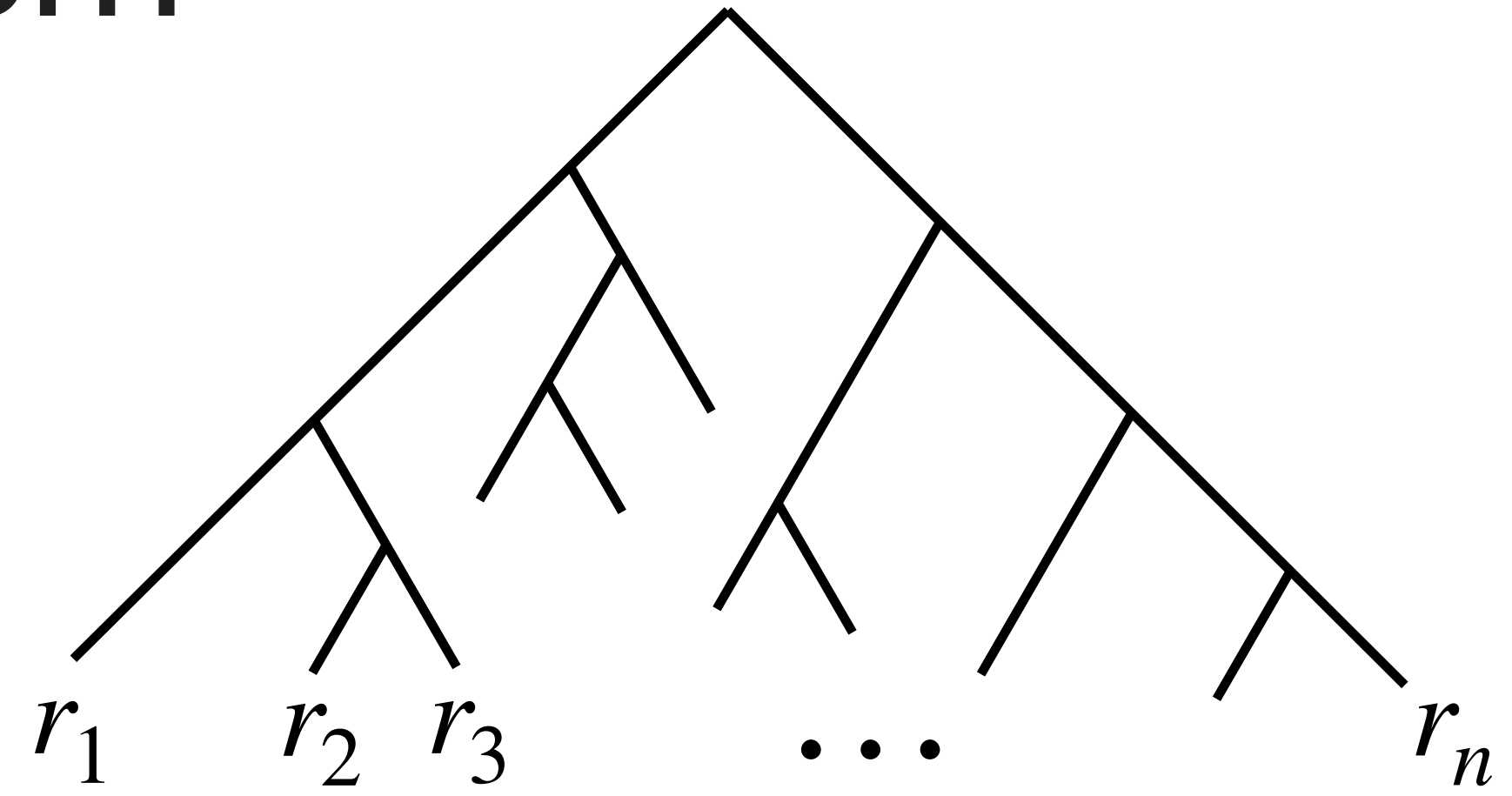
Placement Tree T^*

$$d = D_{T^*} \cdot p$$

DecoDiPhy Problem

$$T^*, p^* =$$

$$\operatorname{argmax}_{T \in \mathcal{T}(T_R, k), p \in \Delta^{k-1}} \|\hat{d} - D_T \cdot p\|_2^2$$

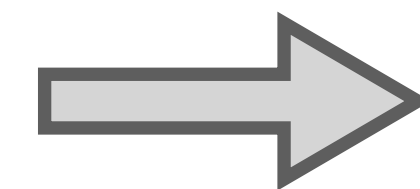
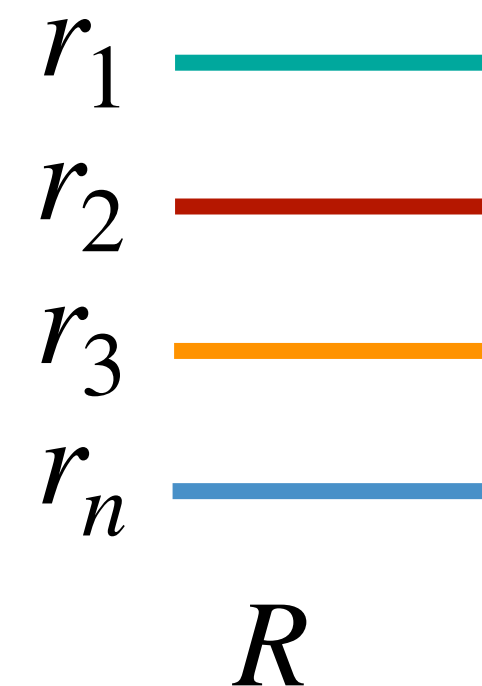
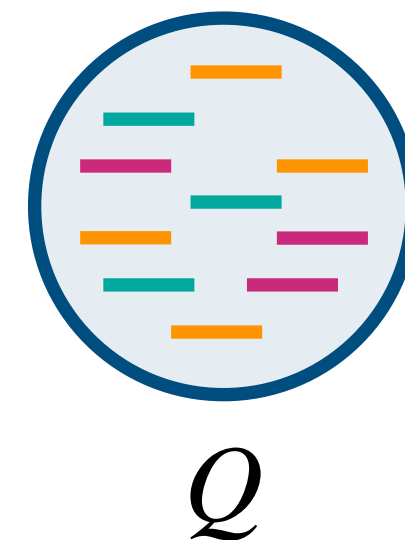


Phylogenetic Tree T_R

Where:

$\mathcal{T}(T_R, k)$ = Space of all trees resulting from placement of k leaves on T_R $O(n^k)$

$\hat{d}$ = Estimated distance vector (Input data)

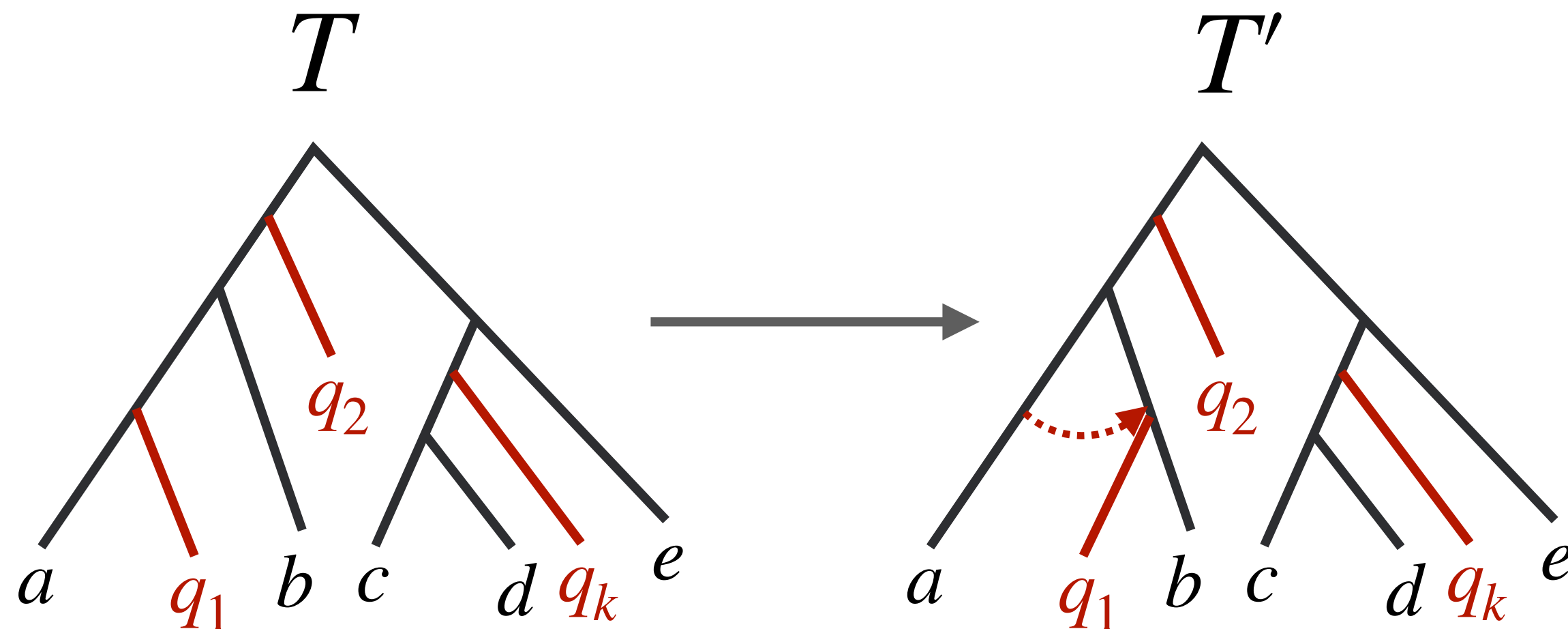
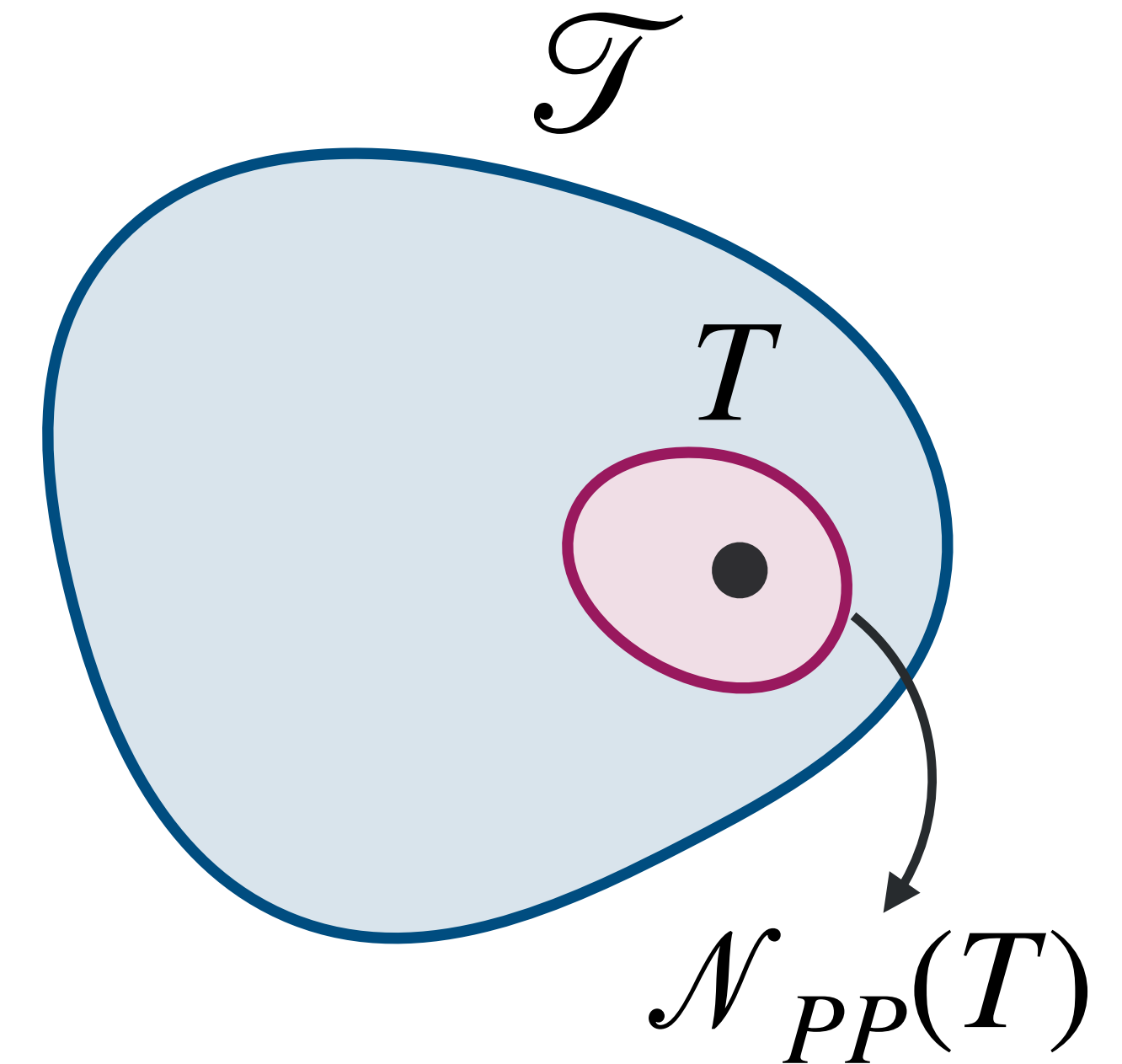


$\hat{d}$

Optimal k –Placement

$$T^*, p^* = \operatorname{argmax}_{T' \in \mathcal{N}_{PP}(T), p \in \Delta^{k-1}} \|\hat{d} - D_{T'} \cdot p\|_2^2$$

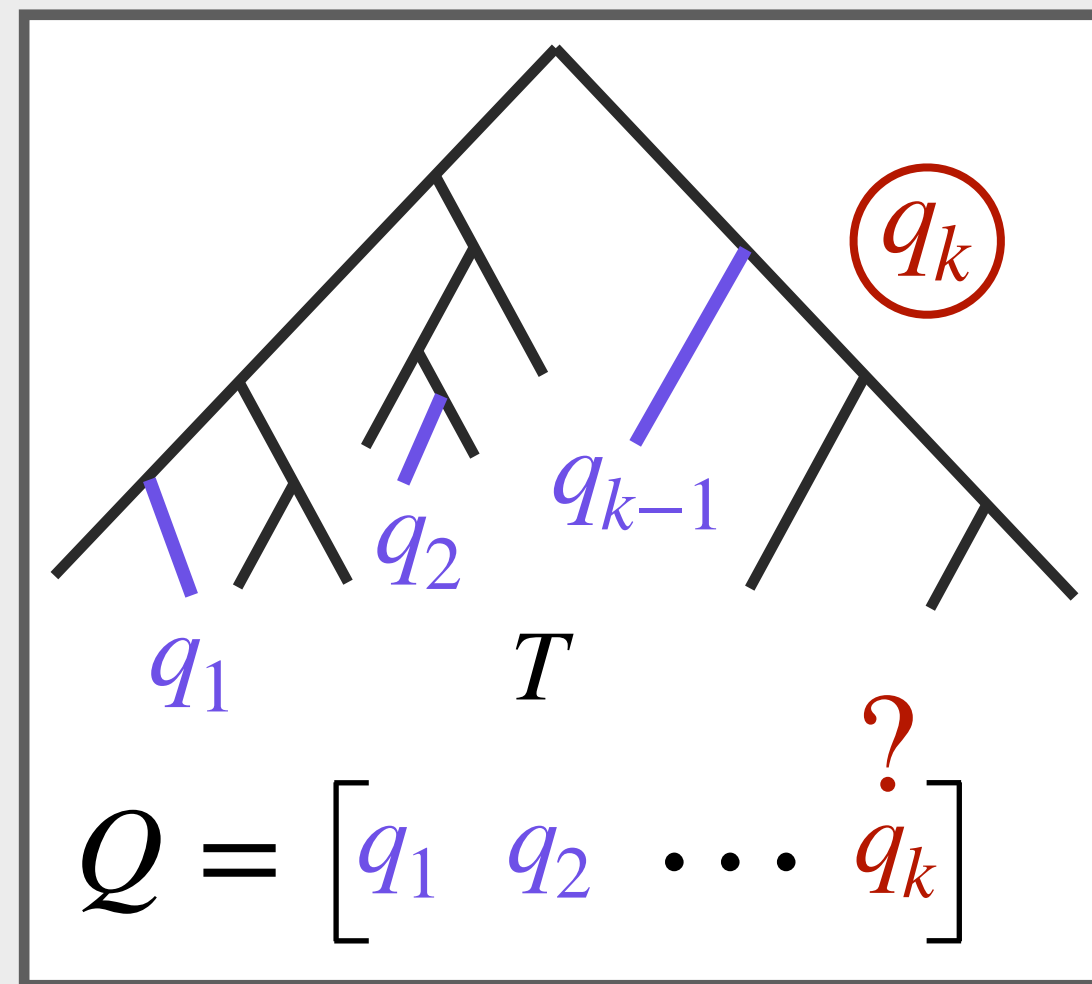
$\mathcal{N}_{PP}(T) = \{T' : T' \text{ differs from } T \text{ in placement of one of } k \text{ queries}\}$



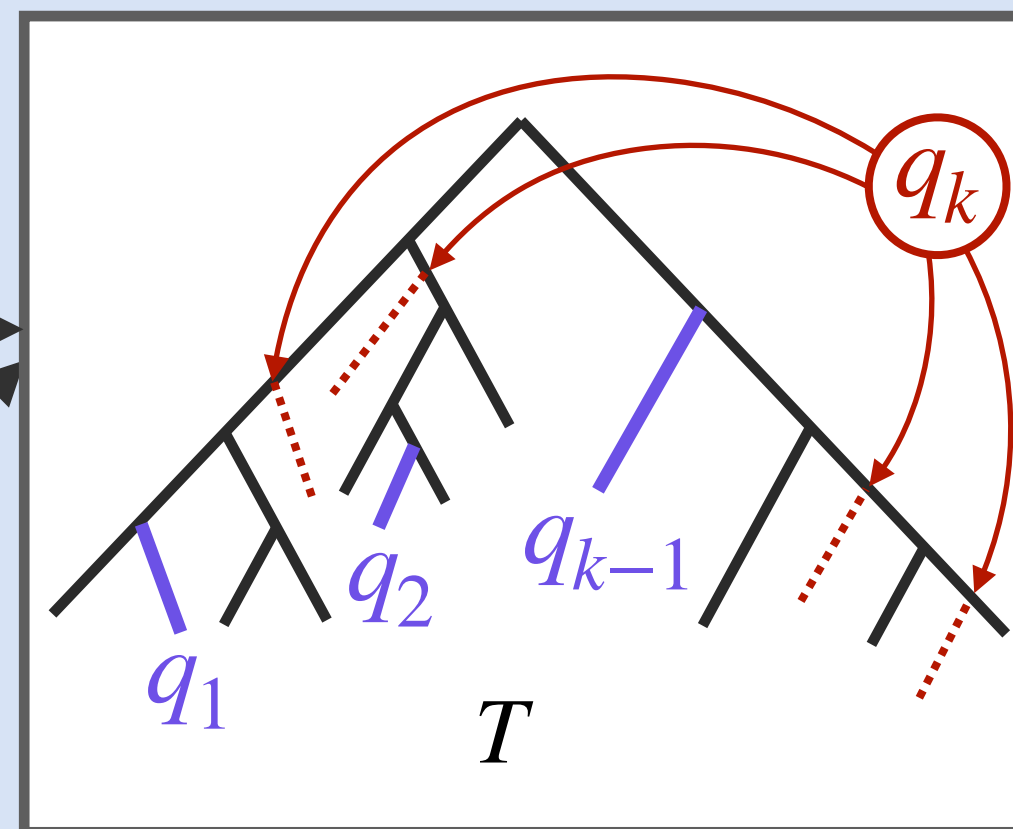
$$|\mathcal{N}_{PP}(T)| = O(kn)$$

Search Algorithm: Greedy + Local Refinements

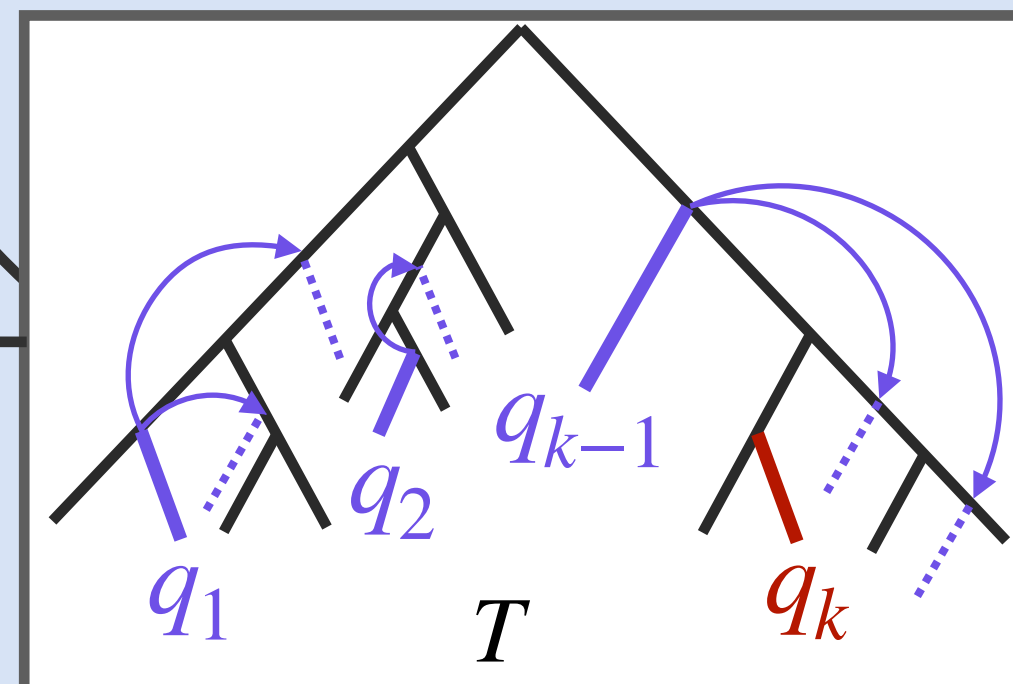
Adding a new placement q_k



Optimal placement transformation of q_k



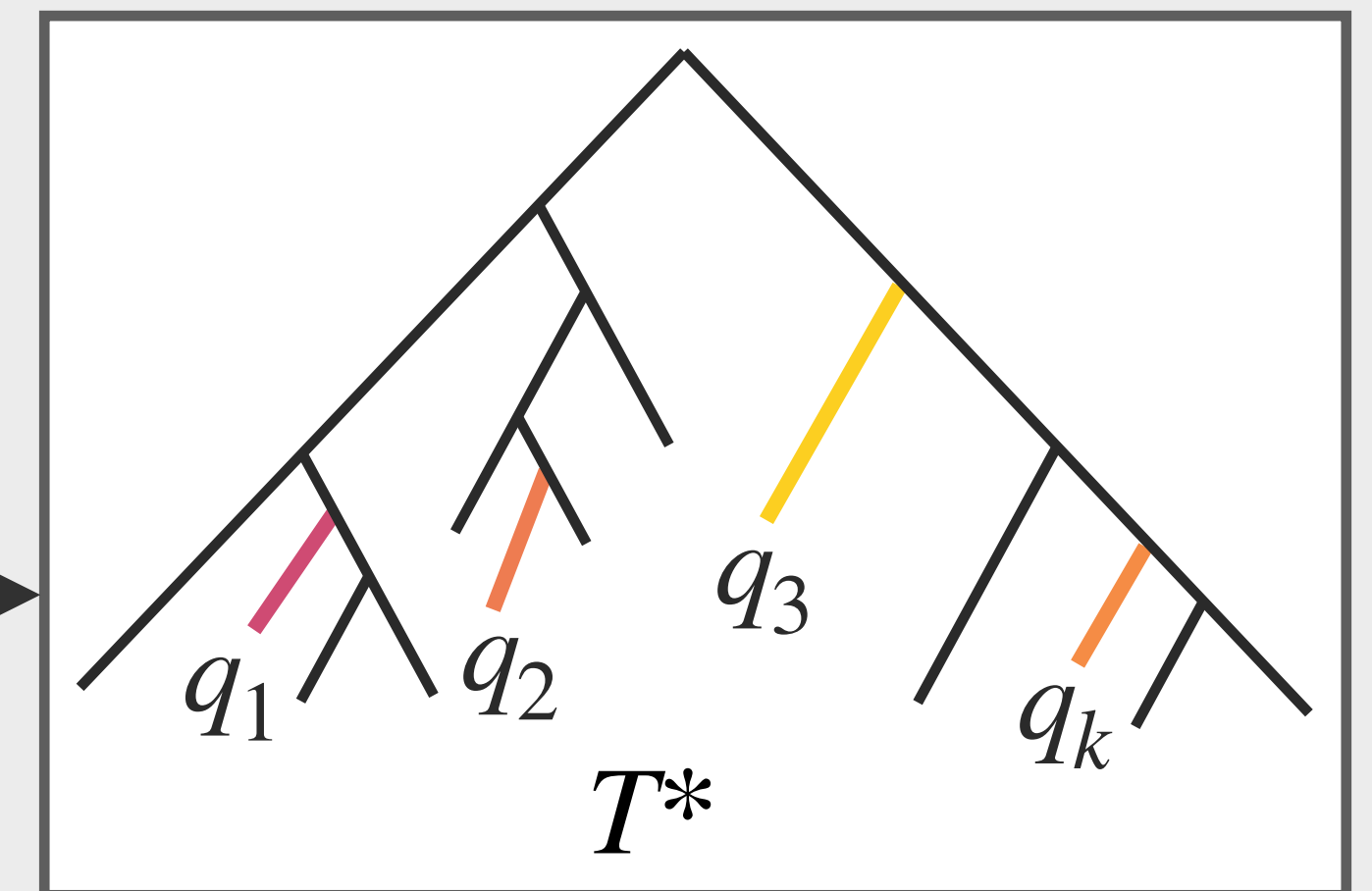
Optimal Placement of $q_1, q_2, \dots, q_{k-1}$



Not converged*

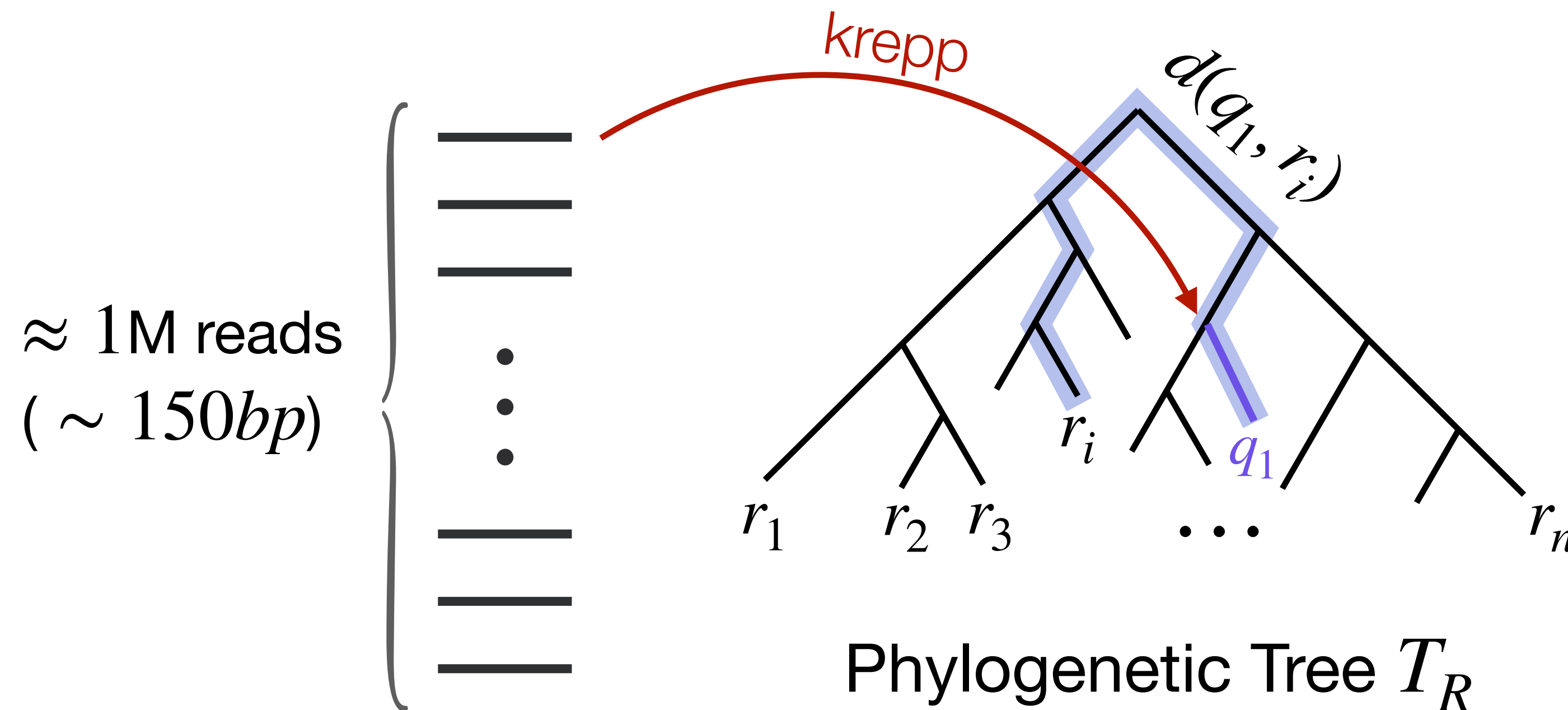
Converged*

Output placements
 $q_1, q_2, \dots, q_k$
with abundances
 $p_1, p_2, \dots, p_k$



Estimating mixture distance vector $\hat{d}$

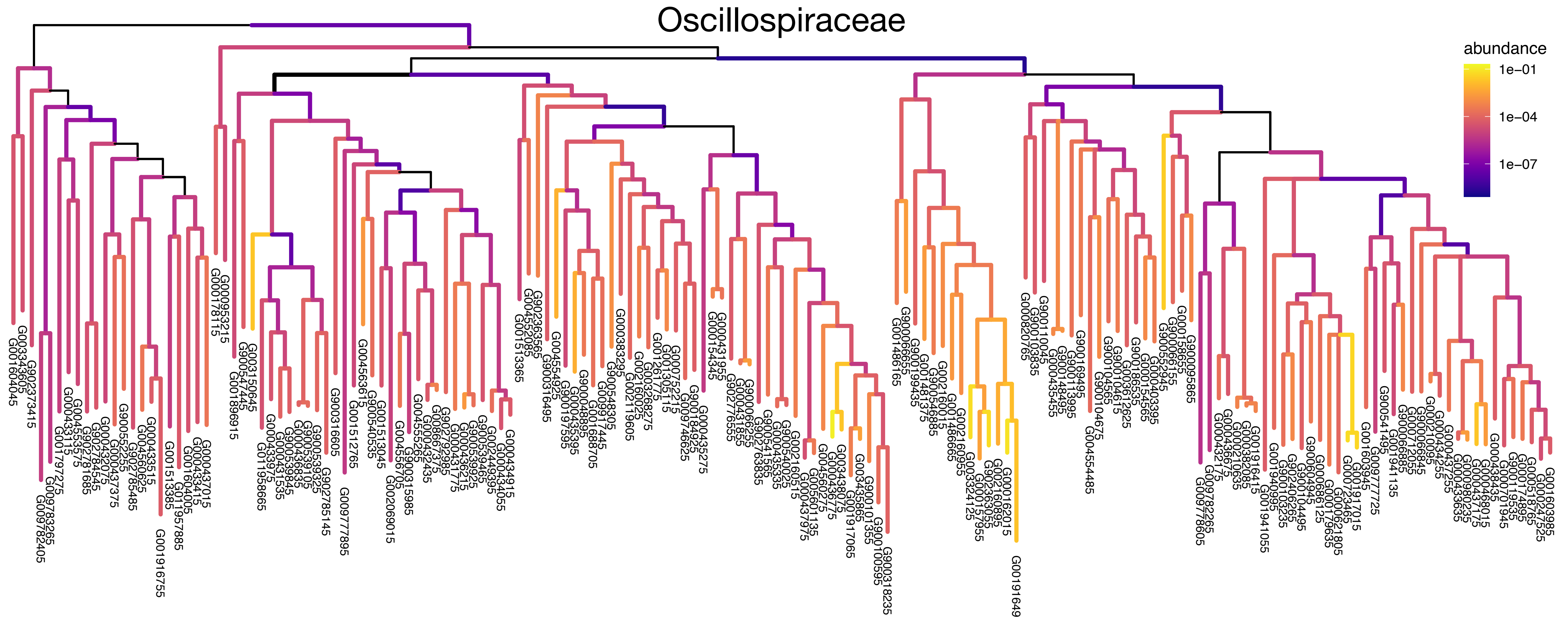
- Average distances can be computed from any read-mapping tool.
- In practice, we use krepp due to high portion of reads mapped.



$$\hat{d}(Q, r_i) = \frac{1}{L} \sum_{j=1}^L d(q_j, r_i)$$

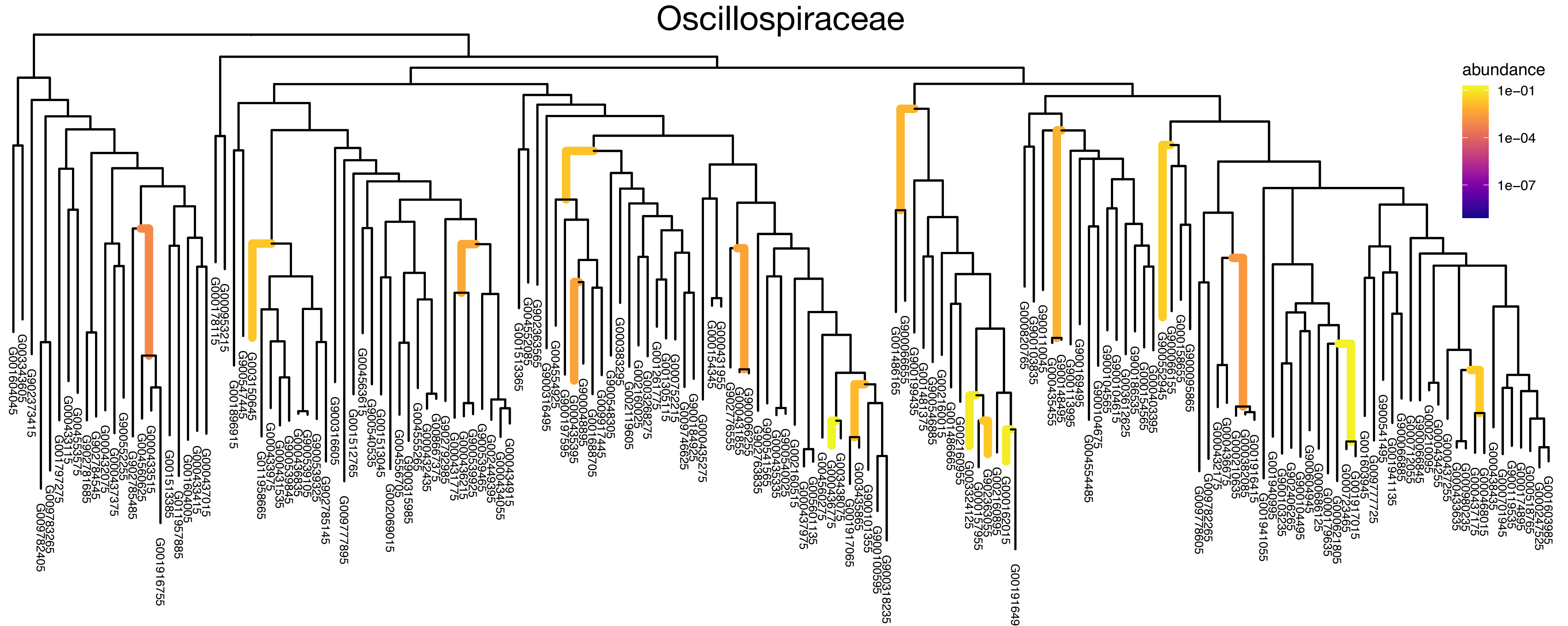
1. Read-level Classification (krepp)

- Inflammatory Bowel Disease (IBD) sample from Franzosa et al. 2019



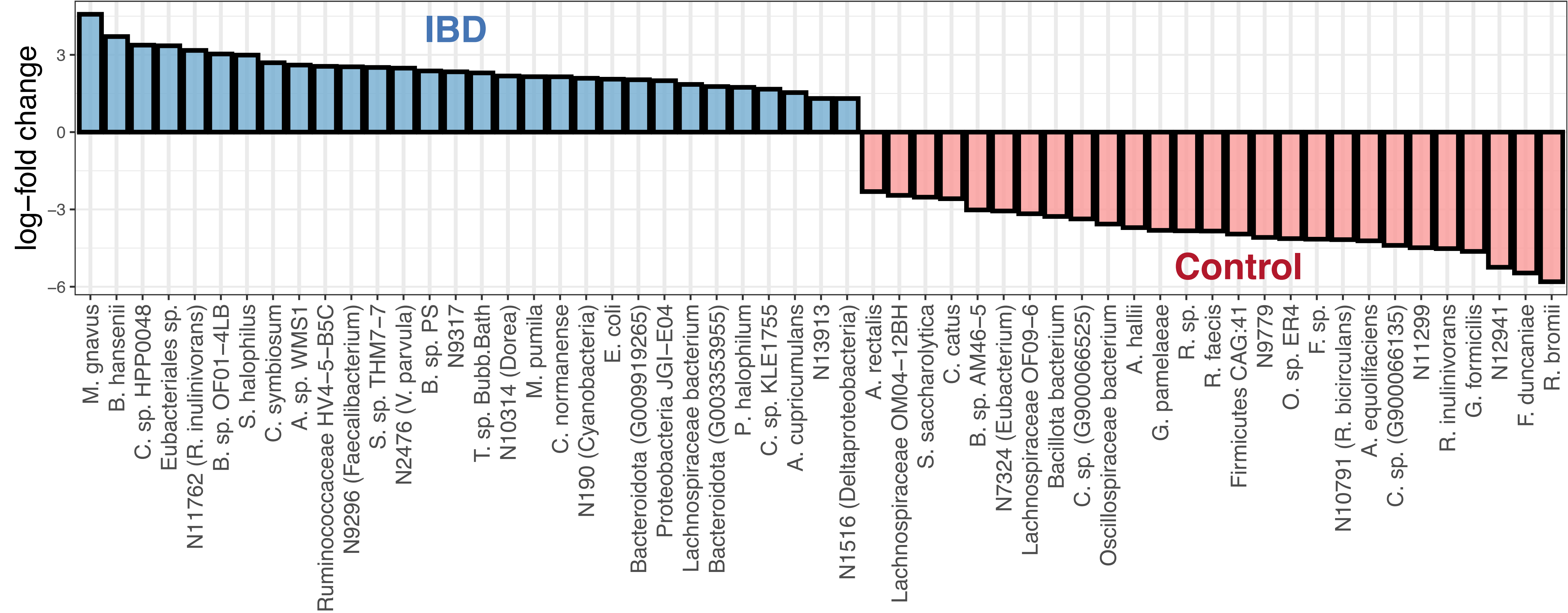
DecoDiPhy

- Inflammatory Bowel Disease (IBD) sample from Franzosa et al. 2019



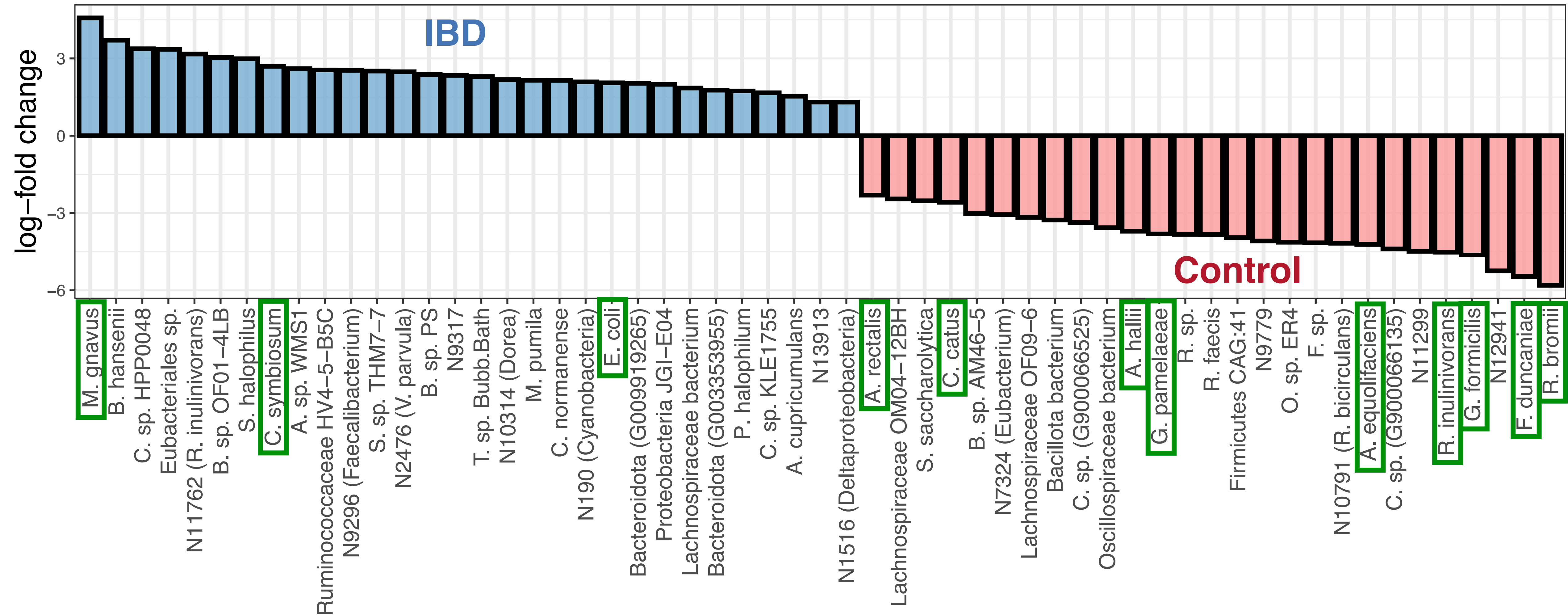
Differentially Abundant Features Identified by DecoDiPhy

- 57 differentially abundant (DA) features detected using ANCOM-BC test (QIMME).
- krepp identifies ~2000 DA features.



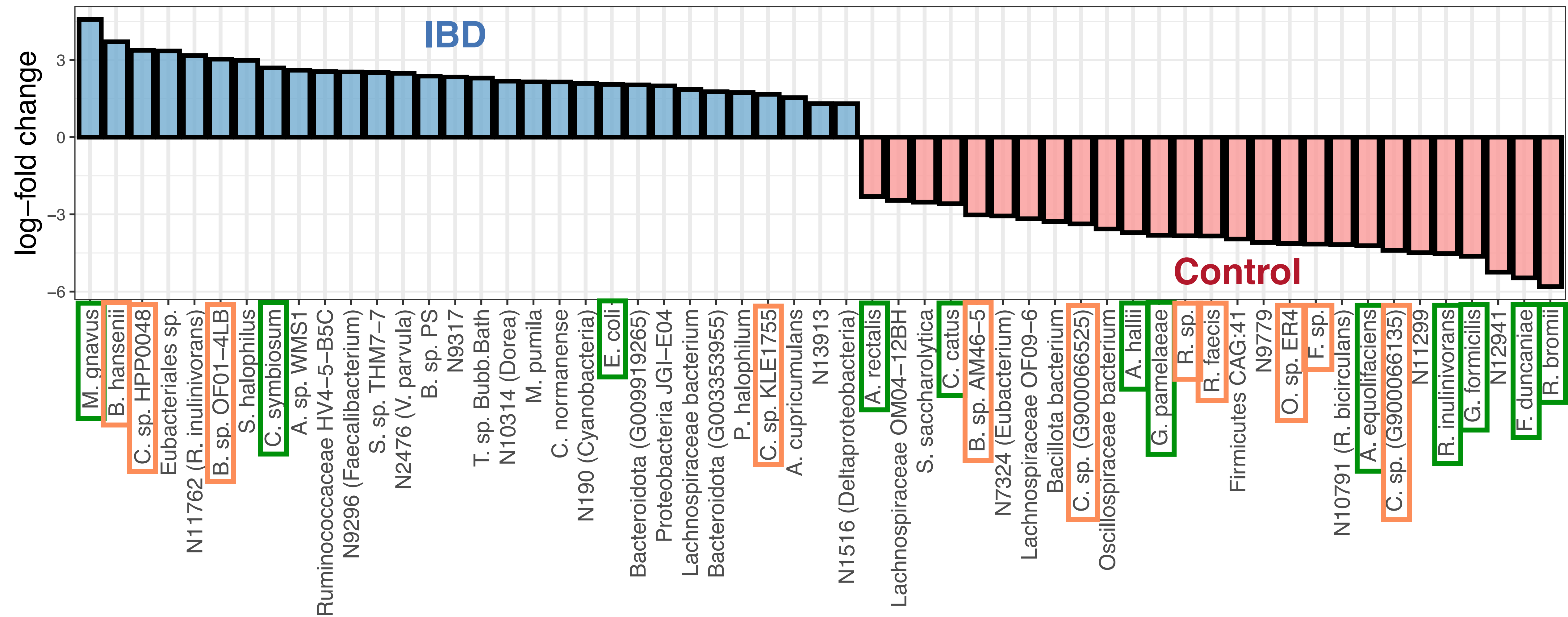
Differentially Abundant Features Identified by DecoDiPhy

- 12 features have known associations with IBD in the literature.



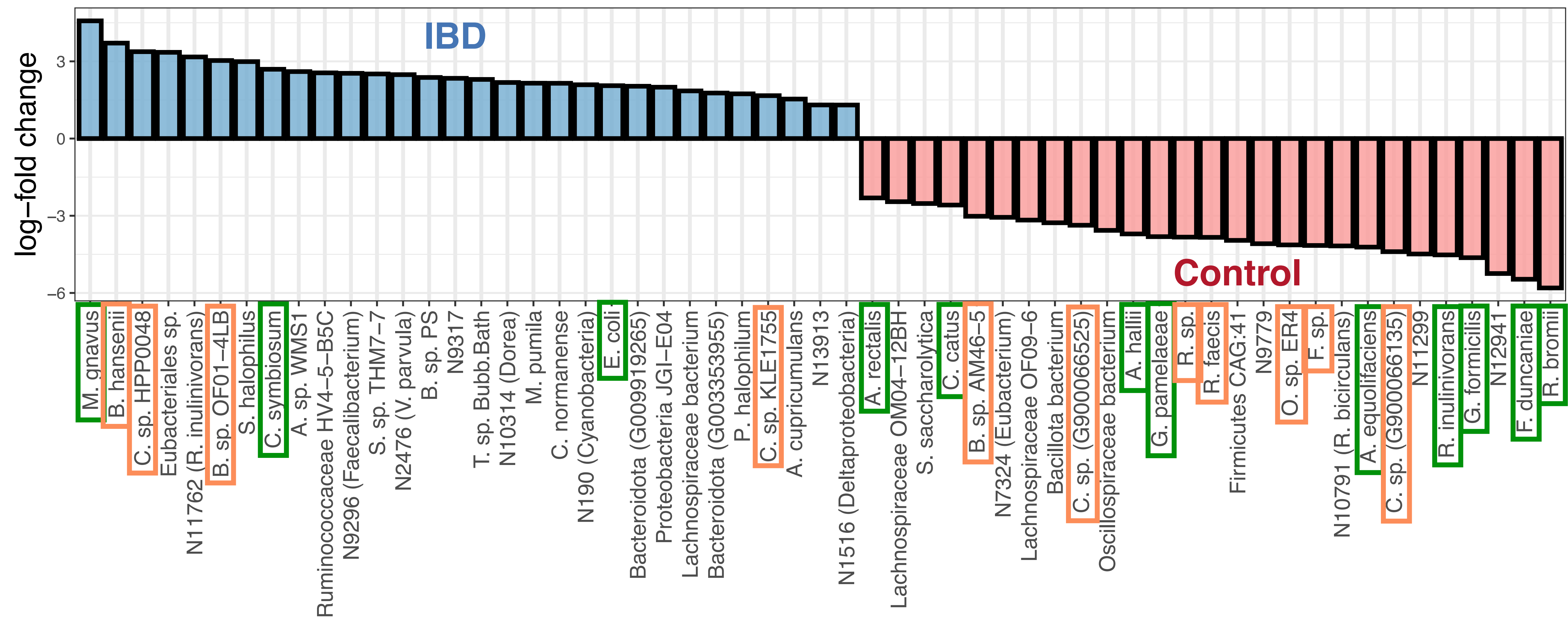
Differentially Abundant Features Identified by DecoDiPhy

- DecoDiPhy detected 11 unclassified/not studied genomes that were also found by other tools (sylph and/or sourmash).



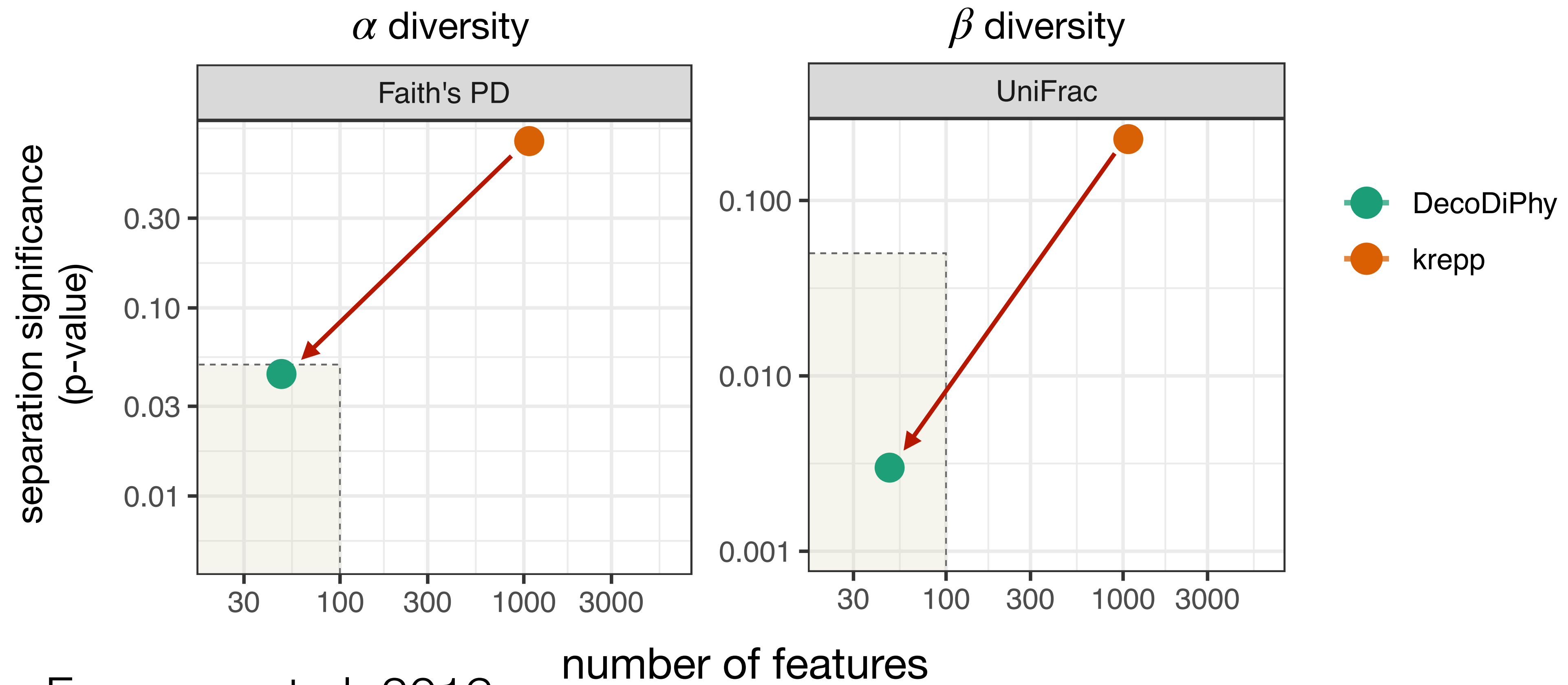
Differentially Abundant Features Identified by DecoDiPhy

- DecoDiPhy also detected 13 internal placements, 11 gut bacteria, and 12 genomes with no gut associations.



DecoDiPhy Features Improve IBD vs Healthy Separation

- Using DecoDiPhy increases separation significance while reducing the number of features

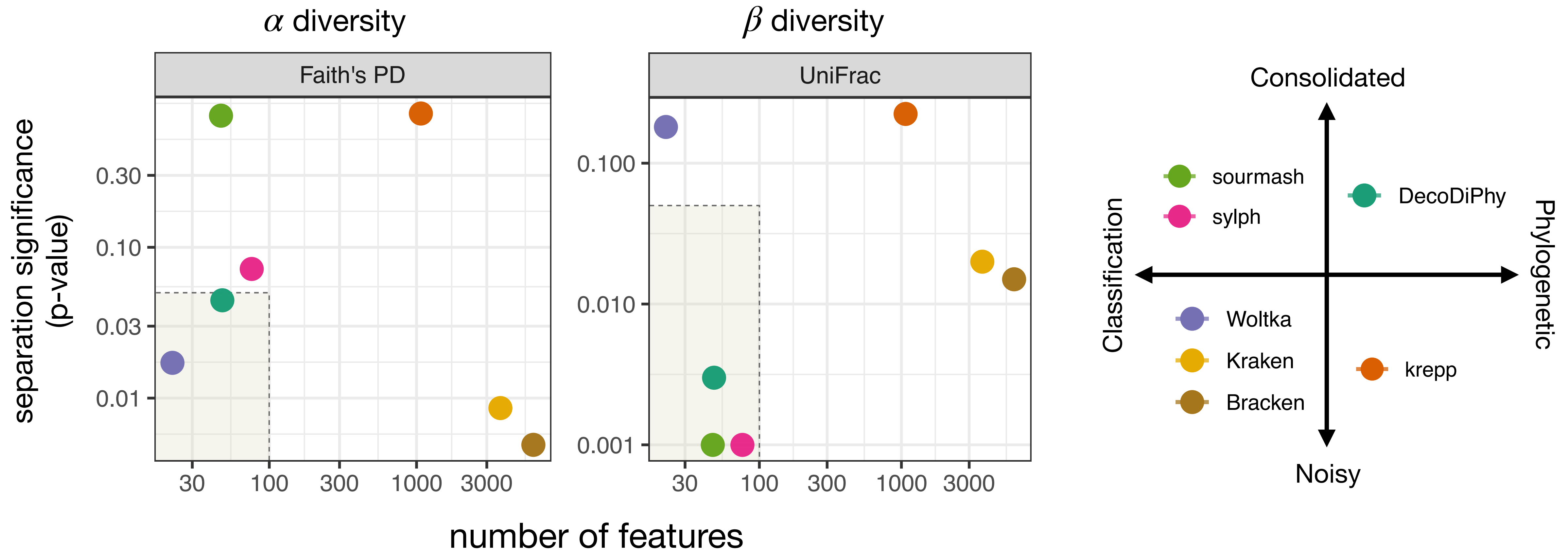


Dataset from Franzosa et al. 2019

220 samples (164 IBD and 56 Control)

DecoDiPhy Features Improve IBD vs Healthy Separation

- DecoDiPhy has strong separation with small number of features

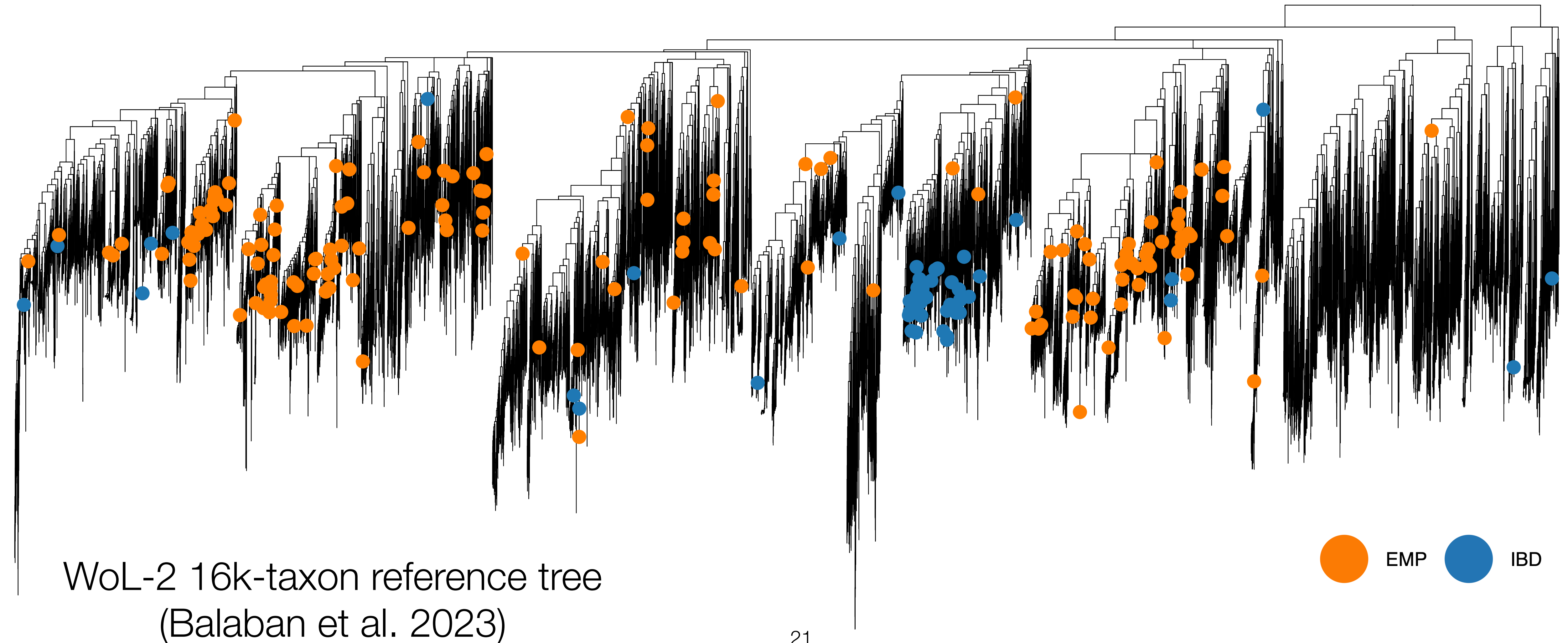


Dataset from Franzosa et al. 2019

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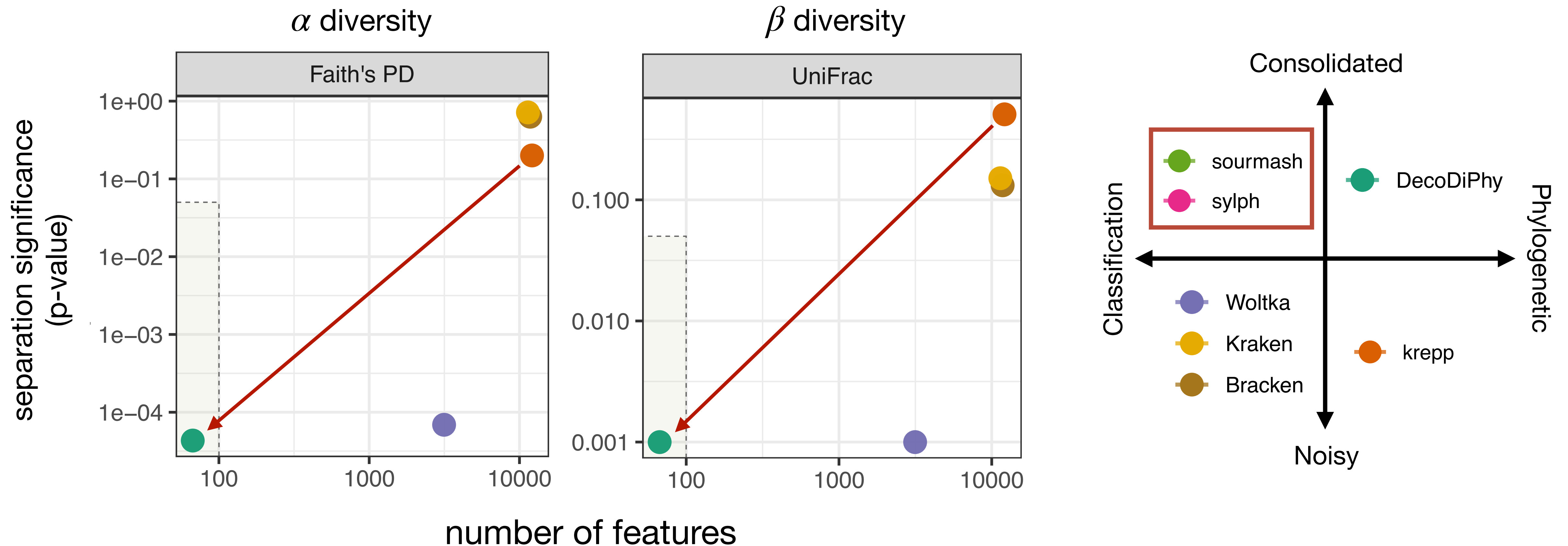
DecoDiPhy Performs Best on High-Novelty Data

- We studied 210 samples from the Earth Microbiome Project (Shaffer et al. 2022)



DecoDiPhy Preforms Best in Saline vs Non-Saline Separation

- sylph and sourmash fail to find any placements for many samples.

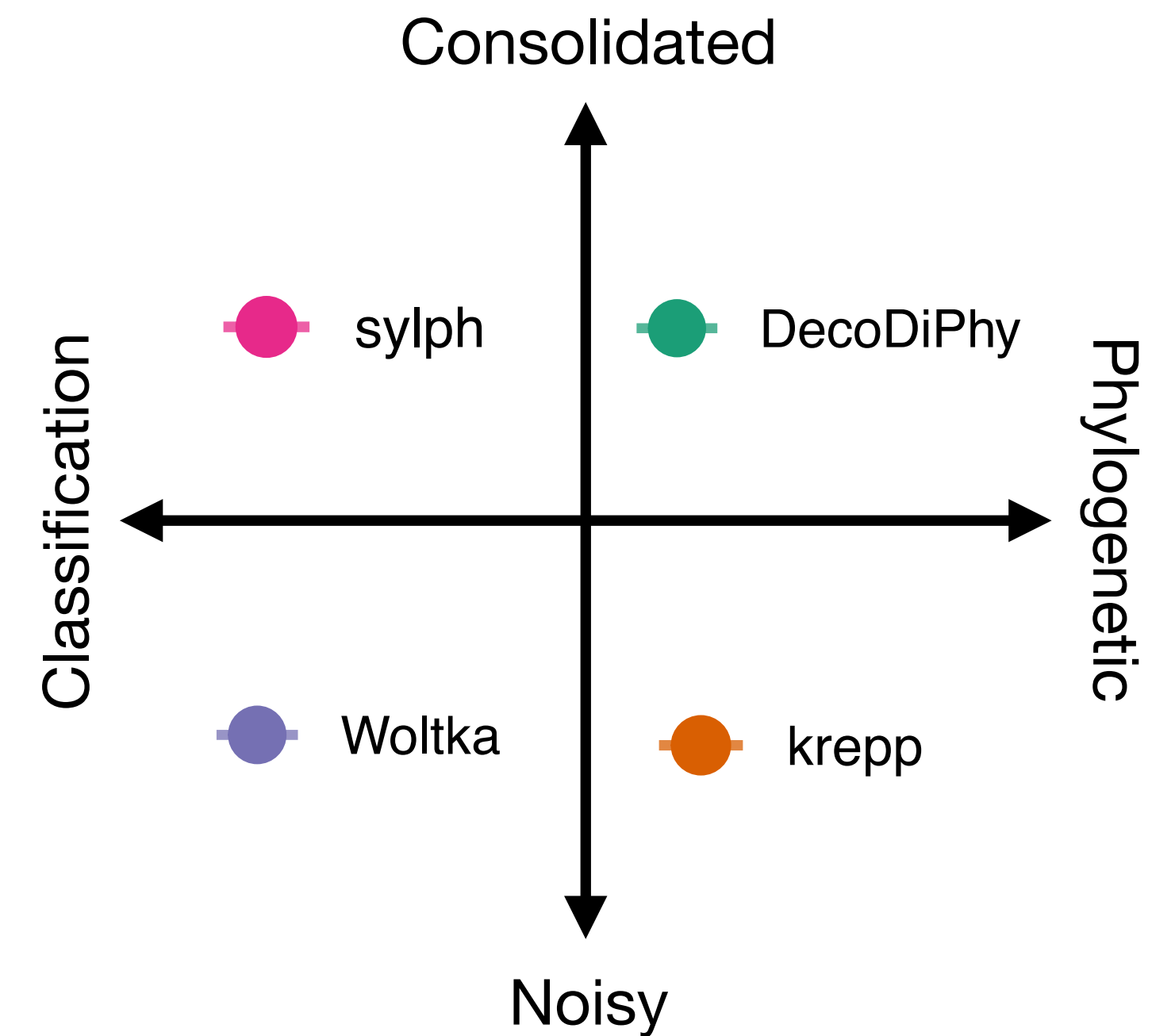
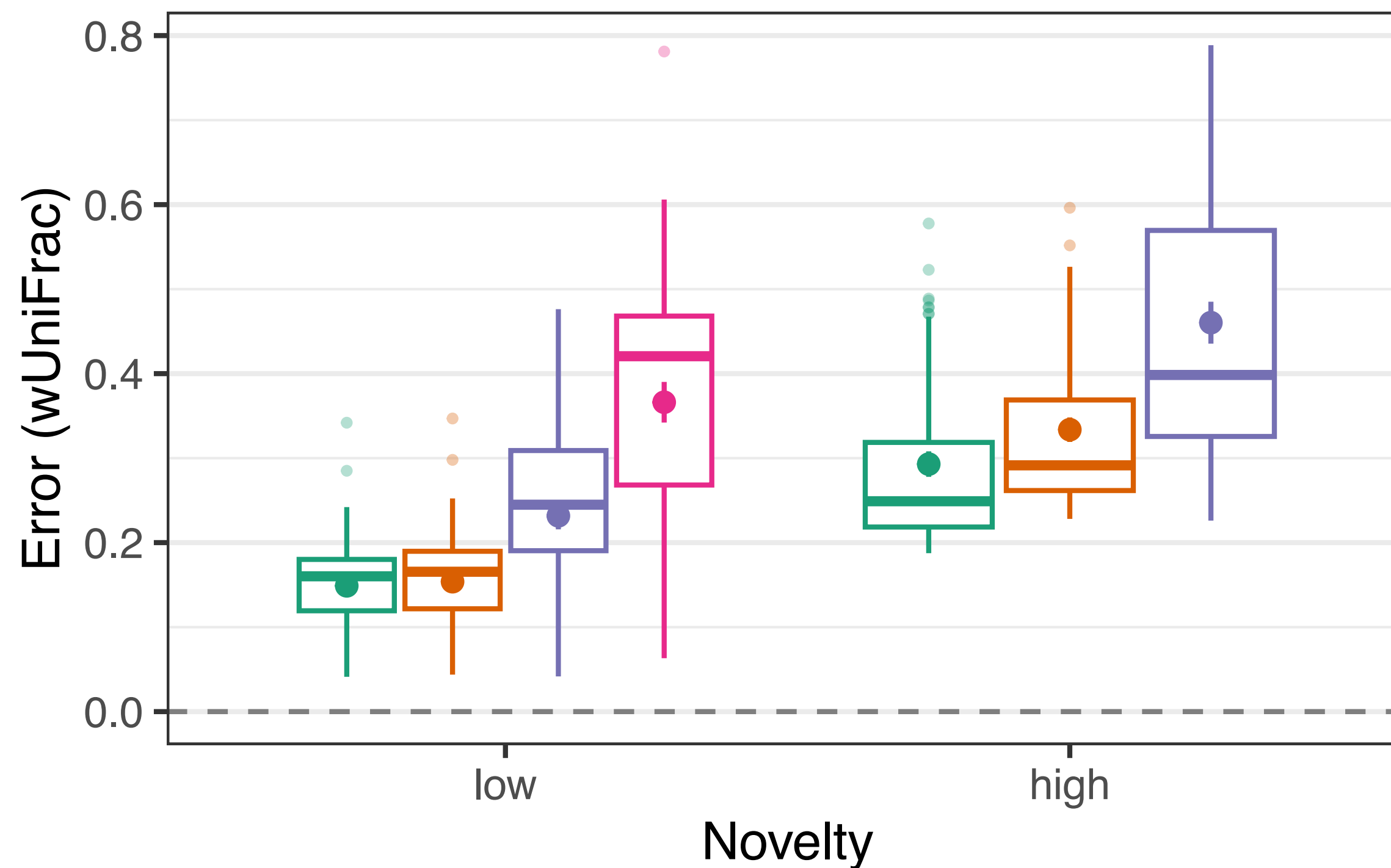


EMP Dataset from Shaffer et al. 2022

210 samples (106 saline and 104 non-saline)

Simulations with Ground Truth

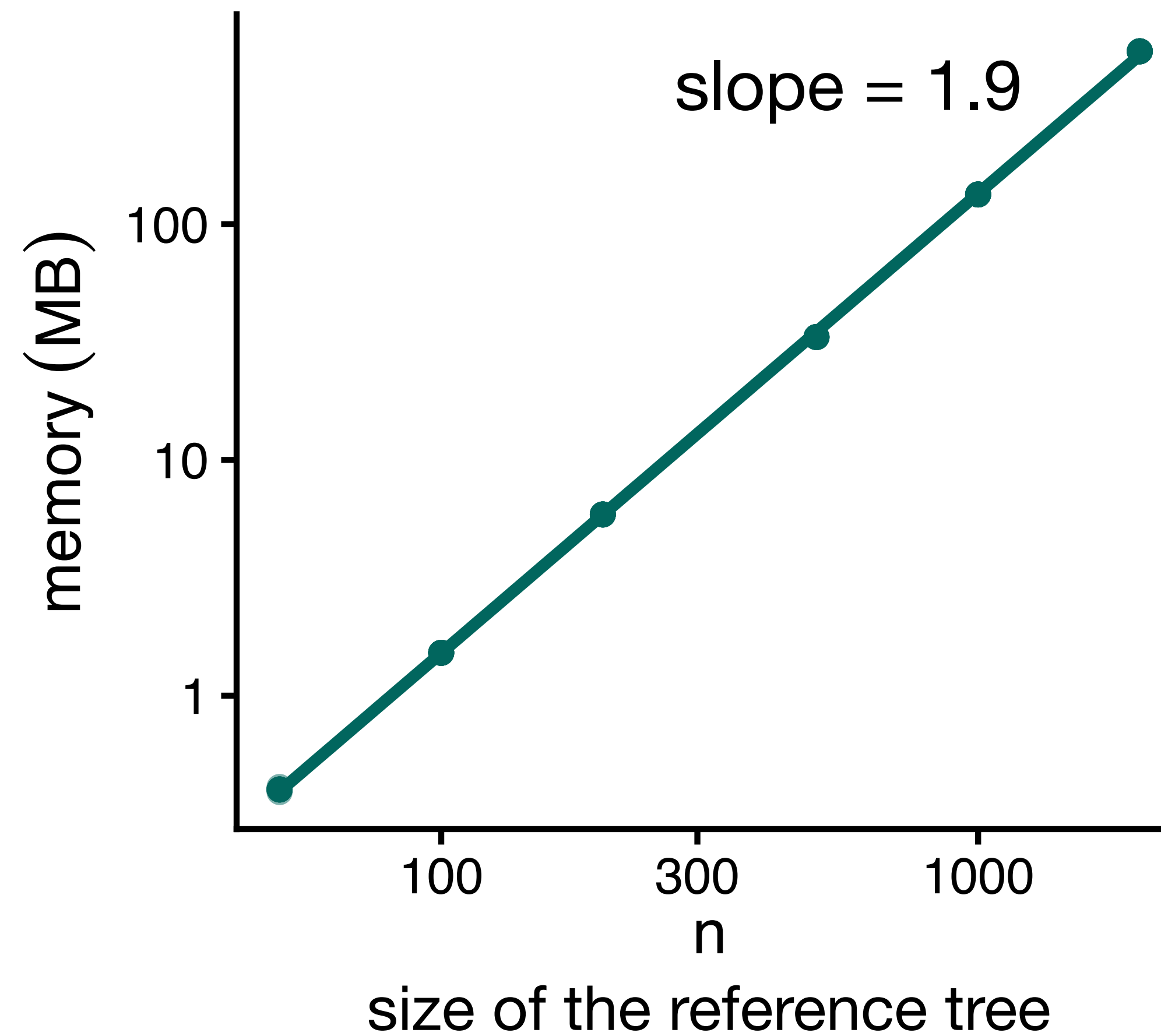
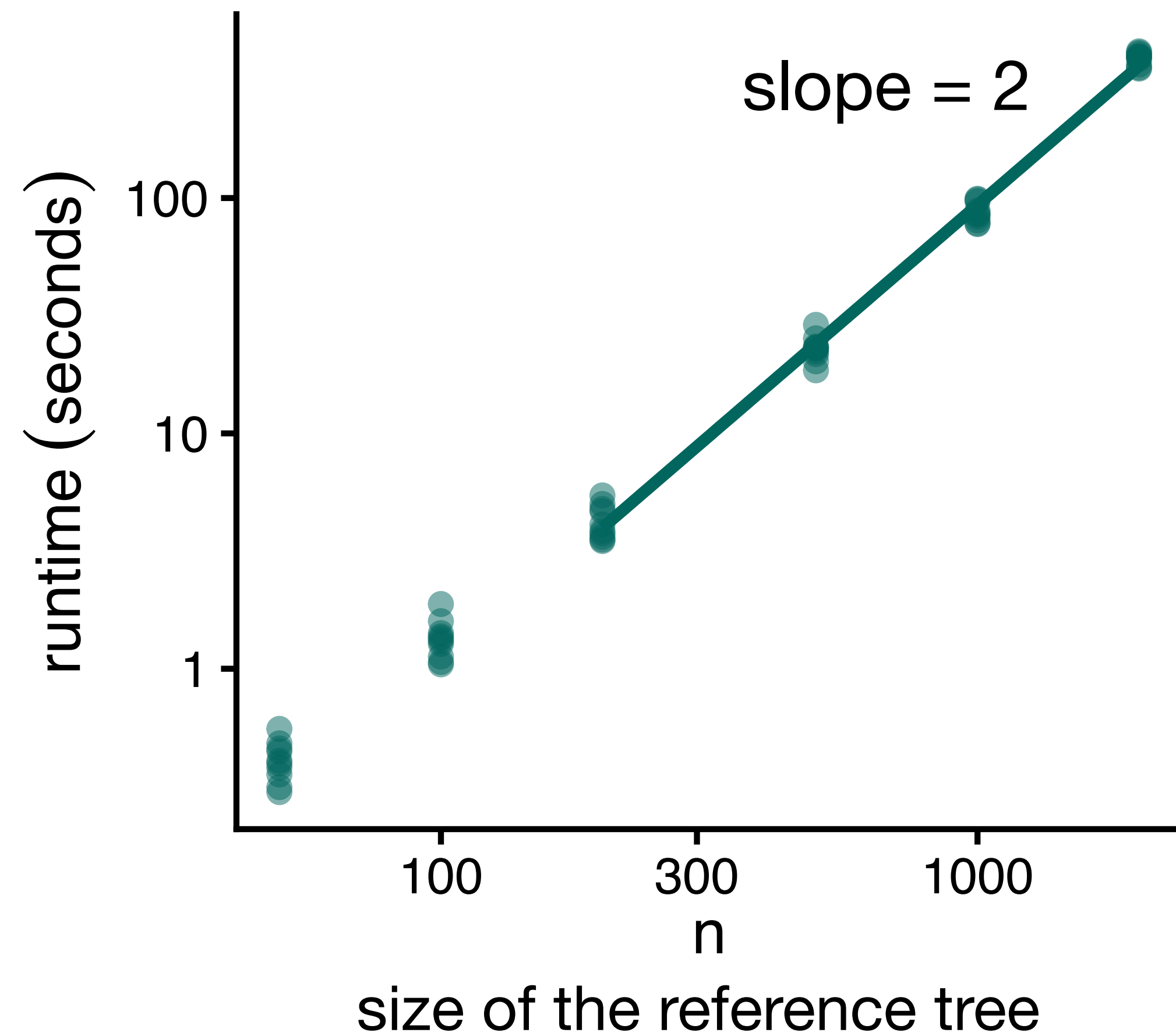
- DecoDiPhy outperforms all methods on simulation dataset, especially in high novelty.
- Sylph fails to detect highly novel taxa.



Simulated Dataset based on the dataset from HMI (Zhu et al. 2022)

Scalability

Total complexity: $O(n^2)$ $k \approx \sqrt{n}$



Summary

- DecoDiPhy is a framework designed to deconvolve metagenomic samples.
- DecoDiPhy uses optimal phylogenetic multi-placements to solve:

$$T^* = \operatorname{argmax}_{T \in \mathcal{T}(T_R, k)} \|\hat{d} - D_T \cdot p\|_2^2$$

- Using DecoDiPhy increases strength and interpretability of the results over read-level classifiers and increases handling of novelty over joint classifiers.